

Lecture 0: Getting Started

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GitHub Education

We'll be using GitHub throughout our course for version control, content delivery, and [Classroom 50](#) for assignment distribution.

To start, create an account on [GitHub](#) and register for a [student/educator discount](#)

- In Week 2 we'll walk through how to sync the course repository using [Github Desktop](#). Feel free to install it now too if you'd like, or wait to decide if it's right for you in Week 2.

Course Website

Speaking of, I will be using this class [Github Repository](#) to distribute all class materials. From here you can access course logistics, and all lecture-related files (i.e. slides and R Markdown files, datasets).

Course Software

Throughout this course we will require a few pieces of (free!) software:

- **R**
- **RStudio**
- Windows: **RTools**
- Mac: **macrtools**

You'll also need to create a **Github Education account**.

R vs. RStudio

- **R** is the **programming language**
- **RStudio** is the **environment** in which we use **R**

Installing R

To install **R**, go to the [R Project website](https://www.R-project.org/).

- **Windows:** "R For Windows > Base > Download R # for Windows"
- **Mac:** "R for macOS > R-#.pkg"
- Where "#" is the current version number



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[R Homepage](#)
[The R Journal](#)

[Software](#)
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[R Binaries](#)

The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux \(Debian, Fedora/Redhat, Ubuntu\)](#)
- [Download R for macOS](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!

- The latest release (2023-06-16, Beagle Scouts) [R-4.3.1 tar.gz](#), read [what's new](#) in the

Installing RStudio

To install **RStudio**, go to the [RStudio Download Page](#)

- Use the links to install RStudio for your operating system.

 RStudio User Guide
Release 2026.08.2

VersionsRelease Notes

Introduction

Getting Started Guide

Accessibility

User Interface

Code

Data

Tools

Productivity

Computational Documents

Deploy

Package Development

Environments

Troubleshooting

Reference

Keyboard Shortcuts

Environment variables

Direct Downloads (Open Source)

RStudio IDE

Direct download links for RStudio Desktop 2026.08.2, the version this documentation describes:

Platform	Download
Windows 11	RStudio-2026.08.2-200.exe
Windows 11 (zip archive)	RStudio-2026.08.2-200.zip
macOS 14+	RStudio-2026.08.2-200.dmg
Ubuntu 22 / 24 / 26, Debian 13	rstudio-2026.08.2-200-amd64.deb
Ubuntu / Debian (tar.gz)	rstudio-2026.08.2-200-amd64-debian.tar.gz
RHEL 9 / 10, Fedora	rstudio-2026.08.2-200-x86_64.rpm
RHEL / Fedora (tar.gz)	rstudio-2026.08.2-200-x86_64-fedora.tar.gz

Windows: Rtools

Rtools is a toolchain bundle that will allow building R packages from source, which we'll need for installing packages that aren't listed on CRAN (i.e. only available direct from GitHub)

- Note: **Rtools** is only for **Windows**!

To install the current version of **Rtools**, navigate to the [Rtools website](#), download the installer, and follow the prompts.

Mac: macrtools

For Mac users, the **macrtools** replaces **Rtools**.

macrtools is a package that installs the R toolchain for mac

To install, run

```
# install remotes if your don't already have it  
if (!require("remotes")) install.packages("remotes")  
  
remotes::install_github("coatless-mac/macrtools")
```

To install the R toolchain, run

```
macrtools::macos_rtools_install()
```